

Distributional Derivatives in One and Two Dimensions

Test Functions and Distributions

- ▶ Let $\Omega \subset \mathbb{R}^n$ be open.
- ▶ The space of *test functions* is

$$C_c^\infty(\Omega) = \{\varphi \in C^\infty(\Omega) : \text{supp } \varphi \Subset \Omega\}.$$

- ▶ A *distribution* on Ω is a linear functional

$$T : C_c^\infty(\Omega) \rightarrow \mathbb{R}$$

that is continuous with respect to the usual test-function topology.

1D Distributional Derivative

Let $\Omega \subset \mathbb{R}$ be open and let $u \in L^1_{\text{loc}}(\Omega)$.

Definition. The *distributional derivative* u' of u is the distribution defined by

$$\langle u', \varphi \rangle := - \int_{\Omega} u(x) \varphi'(x) dx, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Remark. If $u \in C^1(\Omega)$, then u' in the distributional sense coincides with the classical derivative.

Example in One Dimension

Let

$$u(x) = |x| \quad \text{on } \mathbb{R}.$$

- ▶ u is not classically differentiable at $x = 0$.
- ▶ Its distributional derivative is

$$u'(x) = \text{sign}(x),$$

defined via

$$\langle u', \varphi \rangle = - \int_{\mathbb{R}} |x| \varphi'(x) dx.$$

2D Distributional Derivatives

Let $\Omega \subset \mathbb{R}^2$ be open and $u \in L^1_{\text{loc}}(\Omega)$.

Definition. The *distributional partial derivatives* $\partial_x u$ and $\partial_y u$ are defined by

$$\langle \partial_x u, \varphi \rangle := - \int_{\Omega} u(x, y) \frac{\partial \varphi}{\partial x}(x, y) \, dx \, dy,$$

$$\langle \partial_y u, \varphi \rangle := - \int_{\Omega} u(x, y) \frac{\partial \varphi}{\partial y}(x, y) \, dx \, dy,$$

for all $\varphi \in C_c^\infty(\Omega)$.

Gradient in the Distributional Sense

The *distributional gradient* of u is

$$\nabla u := (\partial_x u, \partial_y u).$$

Remark.

- ▶ If $u \in C^1(\Omega)$, the distributional gradient agrees with the classical gradient.
- ▶ This definition allows derivatives of functions with jumps, corners, or weak regularity.

Integration by Parts Interpretation

The definition of distributional derivatives encodes integration by parts:

$$\int_{\Omega} \partial_x u \varphi = - \int_{\Omega} u \partial_x \varphi, \quad \int_{\Omega} \partial_y u \varphi = - \int_{\Omega} u \partial_y \varphi,$$

with no boundary terms, since φ has compact support in Ω .

Connection to Weak Solutions

- ▶ Distributional derivatives are the foundation of
 - ▶ weak derivatives,
 - ▶ Sobolev spaces,
 - ▶ weak solutions of PDEs.
- ▶ A function u is a *weak solution* if it satisfies a PDE after testing against $\varphi \in C_c^\infty(\Omega)$.

The Space $H_0^1(0, \pi)$ and Its Norm

Definition. The Sobolev space $H_0^1(0, \pi)$ is defined as

$$H_0^1(0, \pi) := \overline{C_c^\infty(0, \pi)}^{\|\cdot\|_{H^1}},$$

equivalently,

$$H_0^1(0, \pi) = \{ u \in H^1(0, \pi) : u(0) = u(\pi) = 0 \}.$$

The homogeneous Dirichlet boundary conditions make the Fourier *sine* basis natural.

Fourier sine series. Every $u \in H_0^1(0, \pi)$ admits the expansion

$$u(x) = \sum_{n=1}^{\infty} a_n \sin(nx), \quad a_n = \frac{2}{\pi} \int_0^{\pi} u(x) \sin(nx) dx.$$

Derivative:

$$u'(x) = \sum_{n=1}^{\infty} n a_n \cos(nx).$$

Using orthogonality,

2D Extension: $H_0^1((0, \pi)^2)$

Definition.

$$H_0^1((0, \pi)^2) := \{ u \in H^1((0, \pi)^2) : u|_{\partial(0, \pi)^2} = 0 \}.$$

Double Fourier sine series. Every $u \in H_0^1((0, \pi)^2)$ admits the expansion

$$u(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \sin(mx) \sin(ny).$$

Gradients:

$$\partial_x u = \sum_{m,n} m a_{mn} \cos(mx) \sin(ny), \quad \partial_y u = \sum_{m,n} n a_{mn} \sin(mx) \cos(ny).$$

Using orthogonality,

$$\|u\|_{L^2}^2 = \frac{\pi^2}{4} \sum_{m,n} a_{mn}^2, \quad \|\nabla u\|_{L^2}^2 = \frac{\pi^2}{4} \sum_{m,n} (m^2 + n^2) a_{mn}^2.$$

Sobolev H^1 norm:

$$\|u\|_{H^1((0,\pi)^2)}^2 = \frac{\pi^2}{4} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (1 + m^2 + n^2) a_{mn}^2$$

Thus,

$$u \in H_0^1((0, \pi)^2) \iff \sum_{m,n} (m^2 + n^2) a_{mn}^2 < \infty.$$